

OXIDES

Oxides are compounds of oxygen and other elements. Many oxides are very hard, some have a high specific gravity, and a number are brightly coloured and are sought after as gems. The group includes the chief ores of iron, manganese, aluminium, tin, and chromium. Oxides can occur in hydrothermal veins, igneous and metamorphic rocks, and also, because they can be resistant to weathering and transport, in sands and gravels.

FRANKLINITE

Cubic

H 5½–6 • SG 5.07–5.22

This black or brown zinc iron oxide is found in metamorphosed limestones, notably those at Franklin, New Jersey, USA.

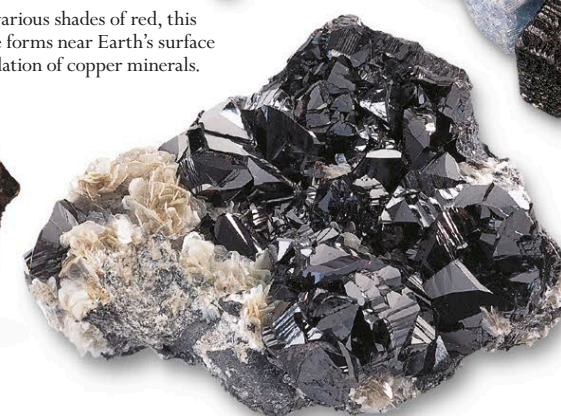
octahedral franklinite crystal



URANINITE

Cubic • H 5–6 • SG 10.63–10.95

This highly radioactive, black- or brown-coloured uranium oxide is the main ore of uranium, which is used in nuclear reactors to produce electricity, and in the construction of nuclear weapons.



CASSITERITE

Tetragonal • H 6–7 • SG 6.98–7.01

Almost the sole source of the world's tin, this tin oxide is mainly found in hydrothermal veins and pegmatites and sometimes as small grains among river gravels.

striated crystal face



vitreous lustre

SAMARSKITE

Monoclinic

H 5–6 • SG 5–5.69

Minerals known as samarskite – a radioactive oxide of various metals including yttrium, iron, tantalum, and niobium – occur in igneous rocks and alluvial sands.



GAHNITE

Cubic • H 7½–8 • SG 4.62

A rare oxide of aluminium and zinc found mainly in metamorphic rocks, gahnite can form dark green or blue to black crystals.



CORUNDUM

Trigonal

H 9 • SG 3.98–4.1

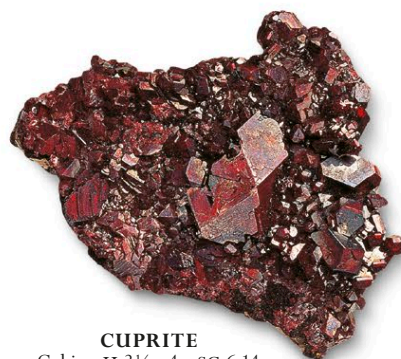
Corundum is an aluminium oxide, second only in hardness to diamond. Ruby (red) and sapphire (usually blue) varieties are used as gemstones.



CHROMITE

Cubic • H 5½ • SG 4.5–4.8

This iron chromium oxide is the only important source of chromium, an element used in making chrome- and stainless-steel.



CUPRITE

Cubic • H 3½–4 • SG 6.14

Found in various shades of red, this copper oxide forms near Earth's surface by the oxidation of copper minerals.



PEROVSKITE

Orthorhombic

H 5½ • SG 3.98–4.26

Discovered in Russia in 1839, this dark-coloured oxide of calcium and titanium forms in igneous and metamorphic rocks.



ILMENITE

Trigonal

H 5–6 • SG 4.68–4.76

This iron titanium oxide is the principal ore of titanium, a high strength, low density metal used in aircraft and rocket construction.

